WORKED SOLUTION

STEP III 2010, Question 13

Probability and Statistics • Correlation of Linear Combinations

STEP III 2010, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s3-q13

This question has a full guided treatment in *Beyond STEP*, Section 13.7, *Manufacturing Correlation*: https://beyondstep.keithmcnulty.org/ch13/correlation.html

The key idea

Everything here is a linear combination of independent variables with mean 0 and variance 1, so every expectation, variance and covariance can be read off from the coefficients: if $U = \sum u_k Z_k$ and $V = \sum v_k Z_k$ then $\mathrm{Cov}(U, V) = \sum u_k v_k$. The question is really building, one variable at a time, three standardised variables with any prescribed correlations, which is the Gram–Schmidt process in disguise. Each new condition pins down one new coefficient.

Solution

Throughout we use bilinearity of covariance, $\mathrm{Var}(Z_k) = 1$, and $\mathrm{Cov}(Z_j, Z_k) = 0$ for $j \neq k$ (which holds because the Z_k are independent). In particular, for constants u_k, v_k,

$$\mathrm{Cov}\Big(\sum_k u_k Z_k,\ \sum_k v_k Z_k\Big) = \sum_{j,k} u_j v_k \,\mathrm{Cov}(Z_j, Z_k) = \sum_k u_k v_k.$$

Correlation of Z_1 and Z_2

Independence gives $\mathrm{Cov}(Z_1, Z_2) = \mathrm{E}(Z_1 Z_2) - \mathrm{E}(Z_1)\mathrm{E}(Z_2) = 0$, so $\mathrm{Corr}(Z_1, Z_2) = 0$.

The variable Y_2

Write $s = \sqrt{1 - \rho_{12}^2}$, which is real and positive since $-1 < \rho_{12} < 1$. Then $Y_2 = \rho_{12} Z_1 + s Z_2$, and by linearity of expectation

$$\mathrm{E}(Y_2) = \rho_{12}\mathrm{E}(Z_1) + s\,\mathrm{E}(Z_2) = 0.$$

By the covariance rule above,

$$\mathrm{Var}(Y_2) = \rho_{12}^2 + s^2 = \rho_{12}^2 + (1-\rho_{12}^2) = 1, \qquad \mathrm{Cov}(Y_1, Y_2) = \mathrm{Cov}(Z_1, \rho_{12}Z_1 + sZ_2) = \rho_{12}.$$

Since $\mathrm{Var}(Y_1) = \mathrm{Var}(Y_2) = 1$, the denominator in the definition is 1, and

$$\mathrm{Corr}(Y_1, Y_2) = \rho_{12}.$$

The variable Y_3

The condition $\mathrm{E}(Y_3) = 0$ holds automatically for every a, b, c, so it tells us nothing. The other three conditions give one equation each. As all of Y_1, Y_2, Y_3 have variance 1, each correlation equals the corresponding covariance.

Correlation with Y_1. $\mathrm{Cov}(Z_1,\ aZ_1 + bZ_2 + cZ_3) = a$, so

$$a = \rho_{13}.$$

Correlation with Y_2. $\mathrm{Cov}(\rho_{12}Z_1 + sZ_2,\ aZ_1 + bZ_2 + cZ_3) = \rho_{12}a + sb$, so $\rho_{12}\rho_{13} + sb = \rho_{23}$ and, since $s \neq 0$,

$$b = \frac{\rho_{23} - \rho_{12}\rho_{13}}{\sqrt{1-\rho_{12}^2}}.$$

Unit variance. $a^2 + b^2 + c^2 = 1$, and since $c \geq 0$ we take the non-negative root:

$$\begin{aligned} c^2 &= 1 - \rho_{13}^2 - \frac{(\rho_{23} - \rho_{12}\rho_{13})^2}{1-\rho_{12}^2} \\ &= \frac{(1-\rho_{12}^2)(1-\rho_{13}^2) - (\rho_{23} - \rho_{12}\rho_{13})^2}{1-\rho_{12}^2} \\ &= \frac{1 - \rho_{12}^2 - \rho_{13}^2 - \rho_{23}^2 + 2\rho_{12}\rho_{13}\rho_{23}}{1-\rho_{12}^2}, \end{aligned}$$

where in the last step the $\rho_{12}^2\rho_{13}^2$ terms cancel. Hence

$$c = \sqrt{\frac{1 - \rho_{12}^2 - \rho_{13}^2 - \rho_{23}^2 + 2\rho_{12}\rho_{13}\rho_{23}}{1-\rho_{12}^2}}.$$

Ask yourself What if the expression under the square root is negative?

Then no real c exists, and there are no random variables at all with these three correlations: the numbers $\rho_{12}, \rho_{13}, \rho_{23}$ are not mutually consistent. For example, $\rho_{12} = \rho_{13} = 0.9$ and $\rho_{23} = -0.9$ is impossible, because if Y_2 and Y_3 are both close to Y_1 they can't be strongly opposed to each other. The question tacitly assumes the given correlations are achievable, which is exactly the condition that this numerator is non-negative.

Final part: the X_i

Take, for $i = 1, 2, 3$,

$$X_i = \mu_i + \sigma_i Y_i,$$

with $\sigma_i > 0$ (we may take σ_i positive, since only σ_i^2 is specified). Then $\mathrm{E}(X_i) = \mu_i + \sigma_i \cdot 0 = \mu_i$ and $\mathrm{Var}(X_i) = \sigma_i^2 \mathrm{Var}(Y_i) = \sigma_i^2$. Adding constants doesn't change covariances, so

$$\mathrm{Corr}(X_i, X_j) = \frac{\sigma_i \sigma_j \,\mathrm{Cov}(Y_i, Y_j)}{\sqrt{\sigma_i^2 \sigma_j^2}} = \frac{\sigma_i \sigma_j}{|\sigma_i \sigma_j|}\,\rho_{ij} = \rho_{ij},$$

using $\sigma_i \sigma_j > 0$. So these X_i have all the required properties.

Checks

- If $\rho_{13} = \rho_{23} = 0$ we get $a = b = 0$, $c = 1$, so $Y_3 = Z_3$, independent of the others, as it should be. If $\rho_{12} = 0$ then $Y_2 = Z_2$ and $a = \rho_{13}$, $b = \rho_{23}$, $c = \sqrt{1 - \rho_{13}^2 - \rho_{23}^2}$, the obvious choice.
- Numerically, with $\rho_{12} = 0.5$, $\rho_{13} = -0.3$, $\rho_{23} = 0.2$ we get $a = -0.3$, $b \approx 0.4041$, $c \approx 0.8641$, and $a^2 + b^2 + c^2 = 1$. A simulation of 2×10^5 triples (using uniform, not normal, Z_k) gave sample correlations 0.500, -0.302, 0.200 for the X_i, with the right means and standard deviations.

Common slips

- Forgetting that $\mathrm{Var}(Y_2) = 1$ means the correlations are just covariances, and then dividing by the wrong thing.
- Writing $X_i = \mu_i + \sigma_i^2 Y_i$: the variance scales by the *square* of the multiplier.
- Taking $\sigma_i < 0$ for one variable, which flips the sign of its correlations.

Beyond STEP

The three conditions say that the matrix A of coefficients (rows $(1, 0, 0)$, $(\rho_{12}, s, 0)$, (a, b, c)) satisfies $AA^{\mathsf{T}} = R$, where R is the correlation matrix. This is the *Cholesky decomposition* of R, and it exists exactly when R is positive semi-definite; the numerator of c^2 is $\det R$. It is how statisticians simulate correlated variables in practice, and with normal Z_k it produces the multivariate normal distribution. Blitzstein and Hwang, *Introduction to Probability*, covers covariance, correlation and the multivariate normal; Strang, *Introduction to Linear Algebra*, explains positive definite matrices and the Cholesky factorisation.

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